Bit- and Power-Allocation Algorithm for Symmetric Operation of DMT-Based DSL Modems

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***Abstract*—Discrete multi-tone (DMT) based digital subscriber loop (DSL) modems must allocate power and information bits to the discrete tones for quadrature amplitude modulation. The bit-allocation (also referred to as bit-loading) algorithms in the literature are designed for modems that operate in the simplex mode in which the upstream and downstream signals use nonoverlapping frequency bands. However, greater data transmission capacity can be achieved by using overlapping frequency bands and echo cancelled duplex operation. The conventional bit-loading algorithms are not well suited for the allocation of data bits for transmission in the duplex mode, and the problem of bit loading for duplex operation has not yet been considered in the literature. This letter introduces an algorithm suitable for echo-canceled duplex DMT modems that use overlapping frequency bands for upstream and downstream transmissions for the purpose of transmitting equal data rates in both directions. Moreover, the algorithm performs the bit loading while jointly minimizing a weighted sum of the power used by the central office and the customer premise transmitters. Simulation results are included to show the higher data rates that are possible with the duplex operation as a function of varying levels of echo return loss enhancement.**

***Index Terms*—Bit loading, digital subscriber loop (DSL), discrete multi-tone (DMT), duplex transmission, echo cancellation, OFDM.**

I. Introduction

THE digital subscriber loop (DSL) modems that use the discrete multi-tone (DMT) technology must use an algorithm for allocating a number of bits to each discrete modulation tone. The literature contains many algorithms [1]–[4] that are designed to allocate the data bits and the budgeted power to the multiple tones. All the algorithms in the literature address frequency-division duplex (FDD or simplex) operation of the modem in which the modem uses nonoverlapping spectra for transmission of upstream (US) signals from the consumer premise (CP) to the central office (CO) and downstream (DS) signals from the CO to the CP. The use of these algorithms is not suitable when frequency-overlapped duplex (FOD or duplex) operation is desired [5]. The ADSL ANSI T1.413 Standard [6] includes an optional duplex mode over the US frequency band with echo cancellation. In such a partial duplex case or in a fully frequency-overlapped DMT-based modem designed for symmetric data rates, the allocation algorithms designed for simplex operation will not yield the highest possible total bi-directional data rate.

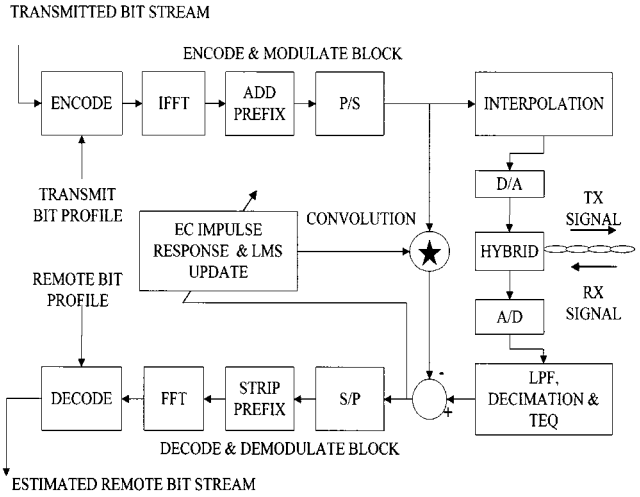


Fig. 1. DMT-based duplex modem.

The formula for Shannon Capacity shows that the theoretical capacity of a channel is directly proportional to the bandwidth of the channel. Therefore, the potential theoretical bi-directional capacity is halved in the simplex mode. As the demand for data rate increases, there is strong incentive to achieve maximum possible data rates. One difficulty in the implementation of the duplex mode is that it is necessary to implement an echo canceler for minimizing the leakage of the transmitted signal into the received signal. In the duplex design, an analog hybrid circuit is used to isolate the two signals (see Fig. 1). Due to loop variations, the hybrid circuit generally cannot provide isolation greater than about 15–20 dB. Depending upon loop length and frequency, the received signal can be attenuated by 20–150 dB. Hence, the leakage of the transmitted signal through the hybrid (the echo) dominates. Therefore, a digital echo canceler (EC) algorithm is required to provide additional isolation. The maximum isolation that can be achieved by the EC is limited by the fixed-point word size in the hardware/firmware implementation of the EC. So, even with the EC, the leakage of some residual echo into the received signal is unavoidable. Such echo leakage causes the CO and the CP power allocation problem to be coupled. Fig. 2 shows the "coupling" that occurs by depicting that the SNR at the CO (CP) is determined by the power transmitted by the CP (CO) and the leakage of the power transmitted by the CO (CP) itself. A number of papers in the literature [7] and [8], for example, describe echo cancellation algorithms for DMT applications; however, these address the problem of EC implementation and not the problem of bit loading with residual echo.

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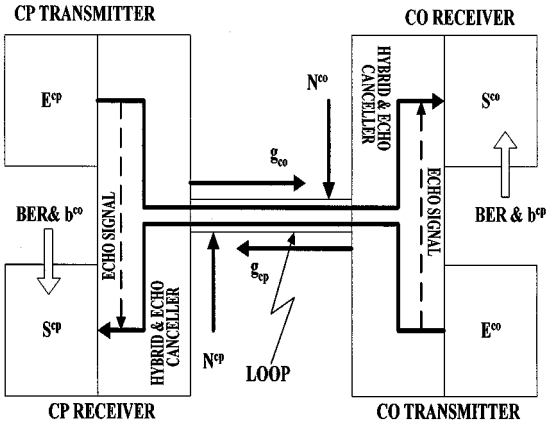


Fig. 2. Duplex allocation is a coupled problem.

Here we describe a bit- and power-allocation algorithm designed to perform allocation of the available power in modems that use overlapping frequency spectra for transmission of equal data rates with identical bit profiles (symmetric) in the US and the DS directions [9]. The allocation is performed within constraints of the power mask, the desired data rate, the desired bit error rate (BER), the coding and modulation scheme to be used, and any other engineering constraints. Section II describes the notation used in the letter and the formulation of the problem. The initialization process is summarized in Section III, and the new algorithm is described in Section IV, followed by the simulation results and conclusions in Section V.

II. Notation and Problem Formulation

The power constraints are given by $\sum_{k=1}^{256} E^{\mathrm{co}}(k) \le P_{\mathrm{Budget}}^{\mathrm{co}}$ and $\sum_{k=1}^{256} E^{\mathrm{cp}}(k) \le P_{\mathrm{Budget}}^{\mathrm{cp}}$, where $P_{\mathrm{Budget}}^{\mathrm{co}}$ and $P_{\mathrm{Budget}}^{\mathrm{cp}}$ are the power budgets for CO and the CP, respectively. The power values allocated to frequency bin k at the CO and the CP are represented by $E^{\mathrm{CO}}(k)$ and $E^{\mathrm{CP}}(k)$, respectively. The power mask constraints per bin are given by $0 \le E_k^{\mathrm{co}} \le M_k^{\mathrm{co}}$ and $0 \le E_k^{\mathrm{cp}} \le M_k^{\mathrm{cp}}$, where M_k^{co} and M_k^{cp} are the maximum power values that can be transmitted in bin k by the CO and the CP transmitters, respectively. The number of bits allocated to frequency bin k at the CO and the CP are denoted by b_k^{co}, and b_k^{cp}, respectively, and each of the two transmitters has a limitation that depends upon the requested data rate ($R_{\max}^{\mathrm{co}}$ and $R_{\max}^{\mathrm{cp}}$), represented by $\sum_{k=1}^{256} b_K^{\mathrm{co}} \le R_{\max}^{\mathrm{co}}$ and $\sum_{k=1}^{256} b_K^{\mathrm{cp}} \le R_{\max}^{\mathrm{cp}}$. Generally, a limit exists on the maximum number of bits that can be allocated to one DMT tone and a limit may exist on the minimum number of bits to be assigned to one bin, $b_{\max}$ and $b_{\min}$ that can be specified as $b_k \le b_{\max}$ and $b_k \ge b_{\min}$ if $b_k \ne 0$. The measured noise power values in bin k at the CO and the CP are denoted by N_{co_k} and N_{cp_k}. The ratio of the residual leakage signal power to the transmitted signal power is referred to as the Echo Return Loss Enhancement (ERLE). The total ERLE in bin k at the CO and the CP, including the total attenuation provided by the hybrid and the EC, is denoted by $\varepsilon_{\mathrm{co}_k}$ and $\varepsilon_{\mathrm{cp}_k}$. Finally, the power attenuation values due to the channel transfer function in bin k as observed at the CO and at the CP, including any coding gain and the equalizer gains at the receiver, are denoted by g_{co_k} and g_{cp_k}. A figure of merit can be defined to maximize the data rates for the CO and the CP within the constraints listed above, while using the minimum possible total power [4]: $J = \sum_{k=1}^{256}[(E_k^{\mathrm{co}} + E_k^{\mathrm{cp}})]$.

III. Initialization

A conventional DMT-based modem that uses separate frequency bands for US and DS signals uses an initialization procedure to determine the channel transfer functions, g_{co_k} and g_{cp_k}, and the noise power spectral densities (PSDs), N_{co_k} and N_{cp_k}. To calculate the optimal allocation profiles for frequency-overlapped operation, it is necessary to determine additional quantities during the initialization period. The first set of these quantities consists of the hybrid transfer functions, $H_{\mathrm{co}}(k)$ and $H_{\mathrm{cp}}(k)$, and the second quantity is the cancellation that the EC is able to achieve. The cancellation is a function of frequency, $\varepsilon(k)$, and the total echo cancellation at the CO and CP can be denoted by $\varepsilon(k)H_{\mathrm{co}}(k)$ and $\varepsilon(k)H_{\mathrm{cp}}(k)$. In this paper, $\varepsilon(k)$ is used as a frequency-independent quantity specified in terms of the ERLE (ε), and the frequency-dependent component of ERLE is incorporated into the hybrid attenuation functions.

The hybrid transfer function is estimated from signals that are recorded during an augmented initialization process. During one phase of the initialization, the CP-transmitter remains quiet, the CO transmits a predetermined pseudorandom noise (PRN) signal, and the CO-receiver records the signal that leaks through the hybrid. The hybrid transfer function is calculated using the known PRN signal and the recorded leakage signal using the same calculation procedure that is normally used to compute the channel transfer function. During the second part of the initialization phase, the CO transmits the PRN signal, and the CO receiver runs the echo canceler. The leakage signal recording is begun after the EC converges. The hybrid transfer function that is calculated from this data would be the overall echo leakage transfer functions, $\varepsilon H_{\mathrm{CO}}(k)$. An analogous procedure is done at the CP to determine $\varepsilon H_{\mathrm{CP}}(k)$. In the subsequent analysis, the hybrid transfer functions H and the ERLE, ε , are used separately in order to demonstrate the effect of the overall cancellation level, ε, on the calculated data rates.

IV. Bit- and Power-Allocation Algorithm

After accounting for the additional "noise" due to the residual leakage through the hybrid and the EC, the SNRs of the received signals at the CO and the CP can be expressed as follows:

$$S_{\mathrm{co}} = \frac{E^{\mathrm{cp}} \cdot g_{\mathrm{co}}}{N_{\mathrm{co}} + E^{\mathrm{co}} \cdot H_{\mathrm{co}} \cdot \varepsilon}$$

and

$$S_{\mathrm{cp}} = \frac{E^{\mathrm{co}} \cdot g_{\mathrm{cp}}}{N_{\mathrm{cp}} + E^{\mathrm{cp}} \cdot H_{\mathrm{cp}} \cdot \varepsilon}. \tag{1}$$

Each term in the equation is a function of frequency bin k, and the index k is dropped for clarity. The SNR that is required at the receiver in order to demodulate the received signal and to

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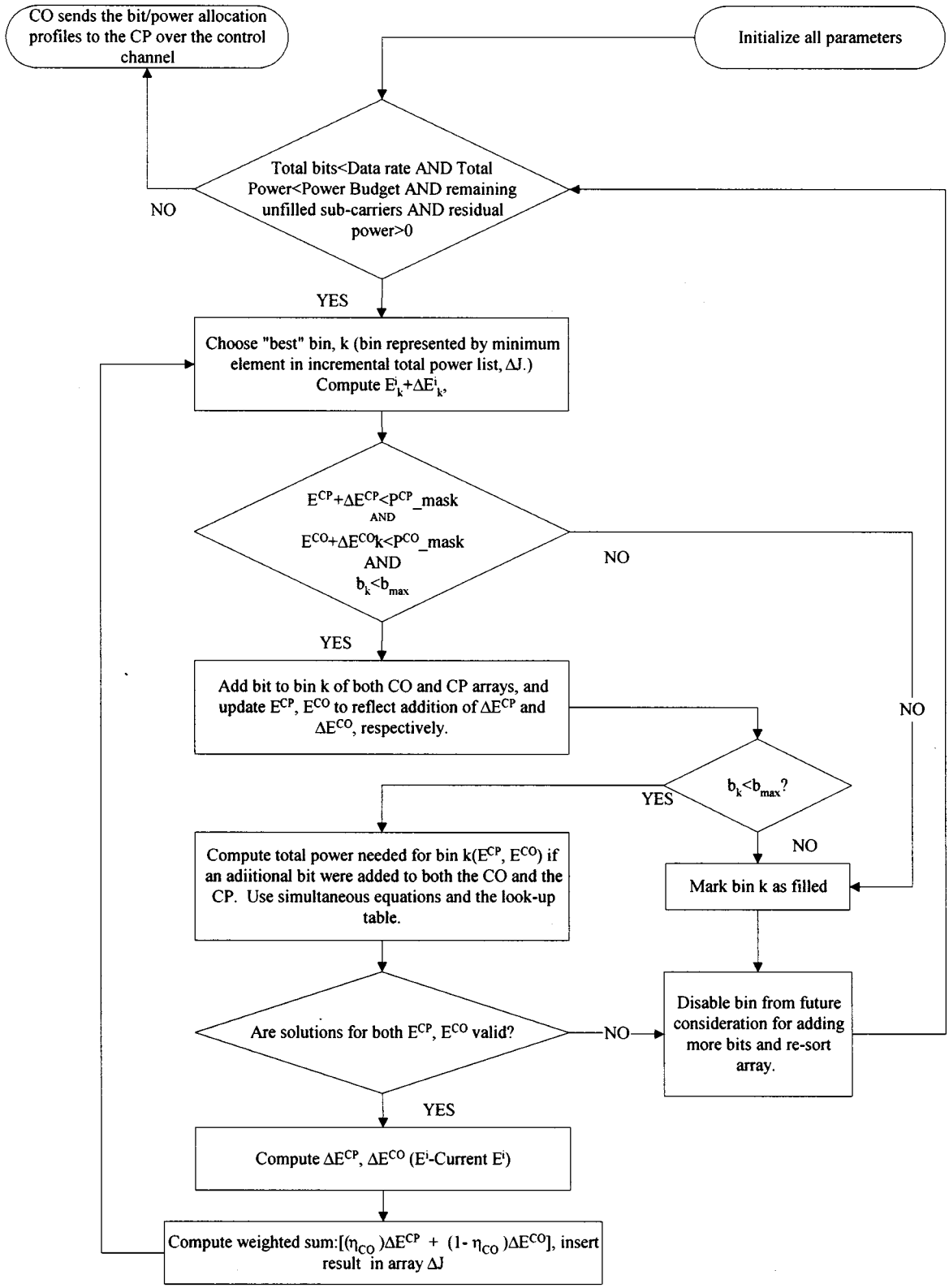


Fig. 3. Bit-loading algorithm.

decode the information bits with the desired maximum bit error rate (BER) is a function of the QAM constellation size. In addition, the modem may also implement a forward error correcting code. Such coding gain can be accounted for if we define and predetermine a two-dimensional look up table, $S(BER,\ b)$, that contains the required receiver SNR for the coding that is to be used, for each possible pair of maximum BERs ($10^{-3} - 10^{-10}$) and number of bits (1–15). If bin k is to be assigned b_k bits, then the SNR required for achieving the desired BER is selected from the table as $S(BER,\ b_k)$. Then the CO and the CP powers can

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be determined by solving the two simultaneous equations (1) in $E^{\rm co}$ and $E^{\rm cp}$

$$E^{\rm cp} = \frac{(g_{\rm cp} \cdot N_{\rm co} \cdot S_{\rm co} + \varepsilon \cdot S_{\rm co} \cdot S_{\rm cp} \cdot N_{\rm cp} \cdot H_{\rm co})}{(g_{\rm co} \cdot g_{\rm cp} - \varepsilon^2 \cdot S_{\rm co} \cdot S_{\rm cp} \cdot H_{\rm co} \cdot H_{\rm cp})},$$
$$E^{\rm co} = \frac{(g_{\rm co} \cdot N_{\rm cp} \cdot S_{\rm cp} + \varepsilon \cdot S_{\rm co} \cdot S_{\rm cp} \cdot N_{\rm co} \cdot H_{\rm cp})}{(g_{\rm co} \cdot g_{\rm cp} - \varepsilon^2 \cdot S_{\rm co} \cdot S_{\rm cp} \cdot H_{\rm co} \cdot H_{\rm cp})}. \quad (2)$$

This solution represents the powers needed at the CP and the CO, such that the specific SNRs (for the selected number of bits, $b^{\rm co}$ and $b^{\rm cp}$) are obtained at the respective receivers for the desired BER. The solution is computed for all bins, such that two vectors of transmit powers are obtained, $E^{\rm co}$ and $E^{\rm cp}$, for all frequency bins $k = 1 \cdots 256$. Given such a solution, a step-by-step algorithm can be designed that loads the bins one bit at a time to the bin that requires the least incremental total power (ΔJ) for one additional bit. To obtain equal data rates in the US and DS directions, the algorithm adds one bit to the CO and to the CP in corresponding bins. Since a large number of binders come together at the CO, it is more important to reduce the CO power than the CP power, to reduce the cross-talk at the CO. Hence, we propose an algorithm in Fig. 3 that uses a modified figure of merit by applying a weight to the incremental CO and the CP powers, $\Delta J = \eta \cdot \Delta E^{\rm co} + (1-\eta) \cdot \Delta E^{\rm cp}$, where $0 \le \eta \le 1$ can be selected to assign the desired relative emphasis to the minimization of the CO and the CP transmit powers.

V. Simulation Results and Conclusion

The duplex algorithm was used to generate the bit- and power-allocation profiles for 26-AWG loops of lengths varying from 3 to 18 kft and for varying levels of ERLE. Fig. 4 is a plot of total bi-directional data rates that can be achieved under duplex operation with various levels of ERLE and of total bi-directional data rates with simplex operation. The analysis was performed for a simulated noise PSD of 10-ISDN, 10-HDSLs in one binder and 4-T1s in adjacent binders. The data rates are calculated for various ERLE values. The simulations show that 60-dB ERLE yields greater data rates than the data rates with simplex operation for all loop lengths. If the ERLE is just 40 dB, then symmetric duplex operation can provide greater rates for loop lengths shorter than 7 kft. For ERLE $\le$ 30 dB, FDD achieves higher total data rates for loops longer than 4.5 kft. Thus, the simulations demonstrate that in order to realize high symmetric data rates with duplex operation, it is necessary to implement an echo canceler that can reduce the transmitted signal leakage into the received signal by at least 50 dB in addition to the isolation provided by the analog hybrid. The advantage of potential wider bandwidth with symmetric duplex operation is lost if the ERLE is poor. In such cases for the proposed symmetric algorithm, the FDD approach provides superior data rates.

In addition to the duplex algorithm, the paper proposes a scheme for minimizing the weighted sum of the CO and CP powers to enable placing greater weight on the reduction of cross talk at the CO. To demonstrate the effect of the weight, a special case of user modems powering on one at a time was simulated, and the allocation algorithm for each modem was constrained to calculate the power needed for a fixed BER and a fixed data rate. The cross-talk caused by the $(n-1)$ modems that are online when the nth modem powers on was added to the WGN level of -140 dBm/Hz. Fig. 5 shows a plot of the total power of the nth modem at the CO to transmit 1.544 Mb/s (T1 rate) on a 9-kft 26-AWG loop for various values of the weight η. As expected, higher weight on the CO power results in lower total transmit power at the CO, which would result in smaller cross-talk in the CO binder.

The algorithm presented in this paper is designed for the special case of *equal data rates and identical bit-allocation profiles* in the US and DS directions. Since bi-directional data rate applications do not necessarily require identical bit-allocation profiles at the CO and the CP, such applications will benefit by the maximization of total bi-directional data rate without the constraint of identical bit profiles in the US and DS directions. Therefore, an area of future work is to design a more general

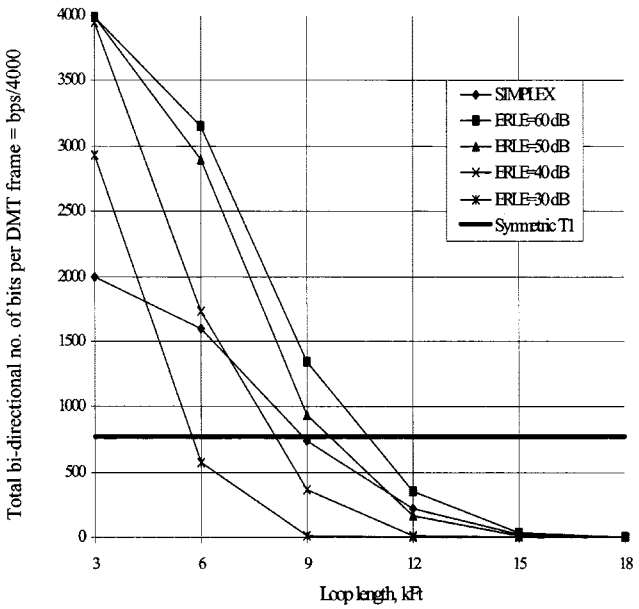


Fig. 4. Comparison of data rate tradeoff with simplex and duplex bit allocation.

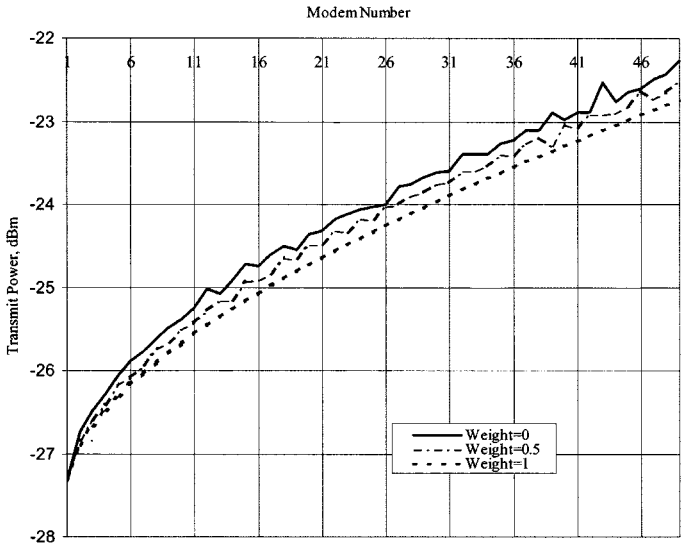


Fig. 5. Effect of weight η on the total transmit power at the CO.

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algorithm for the maximization of total bi-directional data rate with arbitrary US and DS bit-allocation profiles for use in a mode that *permits* frequency-overlapped operation. A general algorithm that is designed to maximize the total bi-directional data rate for a given ERLE would result in data rates that are never worse than simplex operation, because such an algorithm would automatically resort to frequency separation of US and DS signals on loop lengths and in frequency bands where the residual echo is too high. In other regions of operation, such an algorithm would provide higher bi-directional data rates than the conventional simplex operation. The work in this area has recently been completed and is presented in [10] and [11].

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